ISACA Orange County Chapter

Sponsored Problem Challenge (ISACA OC)

Table of Contents

"AI for Everyone: Personal Cyber Safety Coach" ***2***

What is the problem? **2**

Business impact (real-world): **2**

What is the ideal solution? **2**

Minimum viable prototype (MVP) **2**

Your prototype should support at least one input type: 2

And provide three outputs: 2

Stretch goals (bonus) 3

What assets can students use to contextualize it? **3**

A. Datasets (starter options) 3

B. Background context / training resources (Hackathon-provided) 3

C. Constraints (real-world and hackathon-friendly) 3

How ISACA OC will judge "success" (simple rubric you can share) **3**

ISACA Orange County Chapter

“AI for Everyone: Personal Cyber Safety Coach”

What is the problem?

Everyday users (students, families, part-time workers, small clubs) are being hit with phishing, scam messages, and risky links across email/SMS/social platforms. Most people don’t know what to look for, and even when they suspect something is “off,” they don’t have an easy, confidence-building way to decide what to do next.

Business impact (real-world):

- Financial loss (fraud, gift cards, payment redirection)
- Account takeover (school email, banking, social)
- Data exposure (PII, photos, documents)
- Lost time + anxiety + reputational harm for clubs/orgs

Hackathon framing: Build an AI-enabled experience that helps non-experts quickly understand risk and choose a safe next step—without requiring them to be cybersecurity experts.

This aligns with the event’s focus on real-world challenges and AI-enabled prototypes/apps.

What is the ideal solution?

An AI “Cyber Safety Coach” that can:

1. Classify a message/link as likely safe / suspicious / high risk
2. Explain why in plain language (e.g., “urgency + mismatched domain + credential ask”)
3. Recommend next steps (e.g., “don’t click,” “verify via official channel,” “report to IT,” “enable MFA”)
4. Be usable by everyone (accessible language, inclusive design, minimal jargon)

Minimum viable prototype (MVP)

Your prototype should support at least one input type:

- Paste email text OR
- Paste a URL OR
- Upload/screenshot text (optional—only if your team can do it reliably)

And provide three outputs:

- Risk label (Safe / Suspicious / High Risk)
- Top 3 reasons (bulleted, human-friendly)
- Recommended action checklist (what to do next)

Stretch goals (bonus)

- Multi-lingual explanations (English + Spanish, etc.)
- Confidence score + uncertainty ("I'm not sure—here's how to verify safely")
- Privacy mode (auto-redact names/emails/phone numbers before processing)
- "Teach-back" micro-lessons (30-second tips that improve user literacy)
- Lightweight "report button" output (generate a short report a user could forward)

What assets can students use to contextualize it?

You can mix-and-match from the options below depending on your approach (traditional ML, LLM-based classification, rules + AI, etc.).

A. Datasets (starter options)

1. Phishing Email Dataset (Kaggle) – labeled phishing email examples to train/test classifiers.
2. PhishTank – phishing URL intelligence via API and/or downloadable database options (good for link reputation features).
3. Enron Email Corpus – large public email corpus useful for "normal email" language patterns (for baselining/contrast).

Practical note: You do not need a massive model. A strong hackathon solution often wins on user experience + clarity + impact + feasibility, which matches the judging rubric.

B. Background context / training resources (Hackathon-provided)

CPP's hackathon includes training sessions (e.g., prompt engineering, implementation support, storytelling/metrics)—teams should leverage these for a stronger deliverable.

C. Constraints (real-world and hackathon-friendly)

To keep the solution ethical, safe, and "AI for everyone":

- No real personal data required. Use synthetic examples or public datasets.
- Explainability required: your tool must say *why* something is risky.
- Privacy-by-design: avoid storing user inputs; redact where possible.
- Safety: do not provide instructions that enable wrongdoing (e.g., how to craft better phishing).

How ISACA OC will judge "success" (simple rubric you can share)

Aligned to CPP's rubric categories :

- Impact: Would a non-expert actually use it? Would it reduce harm?
- Clarity: Are explanations understandable to "everyone"?
- Feasibility: Could this be piloted at a university/club/SMB?
- Trust: Privacy, transparency, and safe recommendations

- Storytelling: Clear demo narrative in 5 minutes